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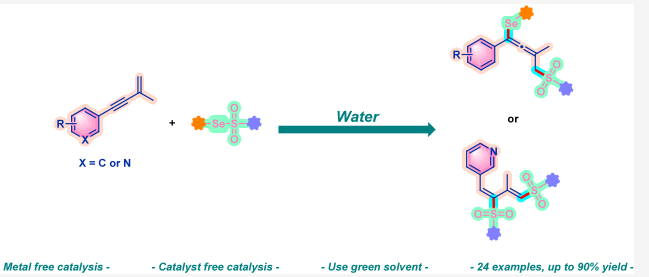
Regioselective 1,4-/1,3-Difunctionalization of 1,3-Enynes with Selenosulfonates in Water

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ABSTRACT: 1,4-/1,3-Regioselective bifunctionalization of 1,3-enynes with selenosulfonates in water under catalyst-free conditions for the construction of sulfonyl allene and 1,3-disulfonyl-conjugated dienes respectively have been developed. The reactions feature mild reaction conditions in aqueous solution and remarkable regioselectivity controlled by substrates.



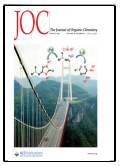
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1. INTRODUCTION

Sulfur-containing compounds are widely found in natural products, pharmaceutical molecules, and functional materials, so their synthesis has attracted much attention.[1] Also, sulfur-containing molecules can serve as important intermediates and ligands for organic synthesis.[2] Selenium (Se), as an important biological trace element, plays a vital role in humans and other higher animals.[3] Allenes are an important component of organic chemistry with a unique cumulative structure.[4] In recent years, the unique structure and properties of allenes have attracted more and more scientists to study them, and the synthesis of allene derivatives has received extensive attention and a variety of synthetic methods have been established.[5] Most of the current methods for constructing allene compounds are to first generate allene radical positive or negative intermediates and then further transform them, but there are relatively few methods for constructing allene compounds based on allene radical intermediates.[6] As a specialized structure in which olefins and alkynes are directly linked, 1,3-enynes have high reactivity and a large number of reactive sites. 1,3-Enynes have been developed as versatile structural units to obtain a variety of highly useful molecular architectures,[7] including alkyne-propyl compounds,[8] biphenyls,[9] and dienes.[10] Based on the specificity of the 1,3-enyne-containing skeleton and our group's research on sulfur and selenium compounds, we envisioned whether we could cleverly design compounds containing the 1,3-enyne skeleton and sulfur and selenium-containing[11] elements to synthesize the connecting skeleton.

Based on the literature research, we found that most of the methods for the construction of allene compounds by 1,3-enynes require the use of metal catalysis. With the further investigation of the allene chemistry and the development of the concept of green chemistry, metal-free catalytic methods for the construction of allene compounds have been developed. In 2019, Ma's[12] group reported a copper-catalyzed 1,4-addition reaction of sulfonyl iodides with 1,3-enynes affording various allenyl halides in high yields under mild conditions has been developed. In the same year, Wang's[13] group reported a nickel catalyst to generate different fluoroalkyl radicals from readily available and structurally diverse fluoroalkyl halides to access 1,4-difunctionalization of 1,3-enynes by a radical relay. In 2020, Bao's[14] group reported the reaction of 1,3-enyne with NFSI for the construction of difluoromethylated allene compounds (Scheme, 1). In 2021, Du's[15] group reported the NHC-catalyzed 1,4-alkylcarbonylation of 1,3-enynes, which introduced difluoromethyl in the alkyne backbone without metal catalysis (Scheme, 1). This method is applicable to a range of alkyl radical precursors and has good regioselectivity and functional group tolerance. In contrast, our group reported a metal-free catalytic selective bifunctionalization reaction of 1,3-enynes with selenosulfonate. Both 1,4-bifunctionalization of 1,3-enynes to construct allene compounds and 1,3-bifunctionalization of 1,3-enynes to construct disulfonyl-conjugated dienes can be achieved by modulation of the substrate.

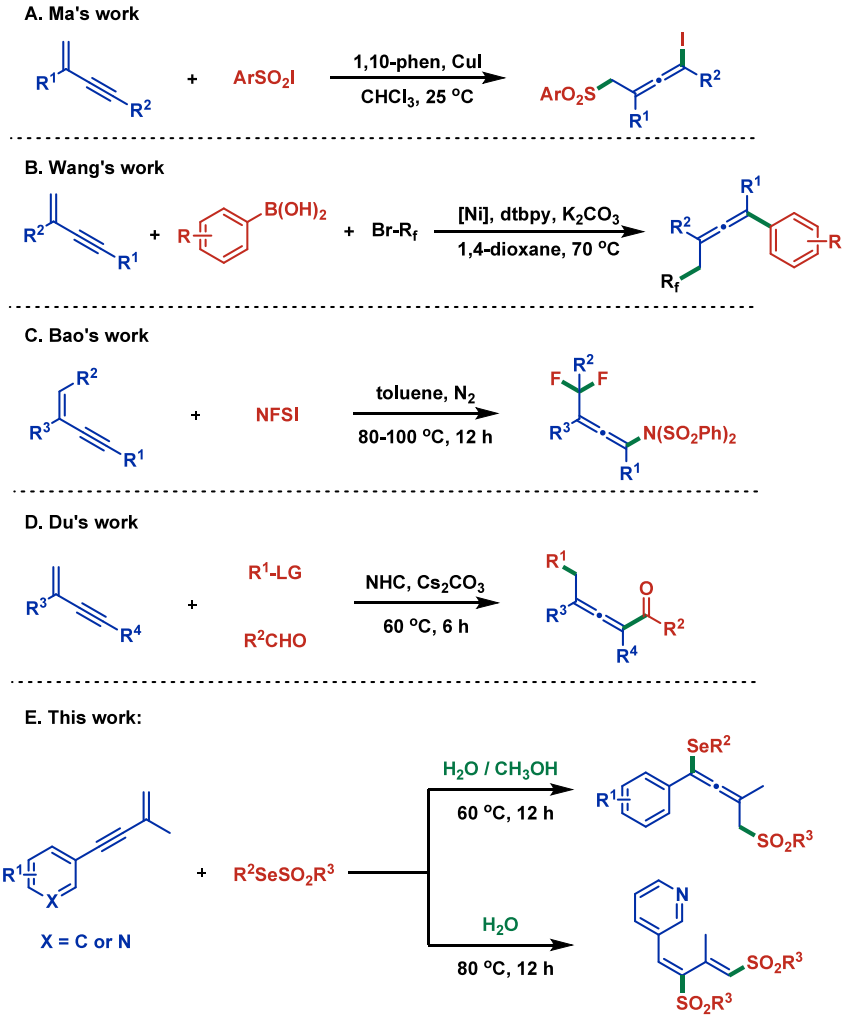
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Scheme 1. Synthesis of Structures Containing Allene Compounds



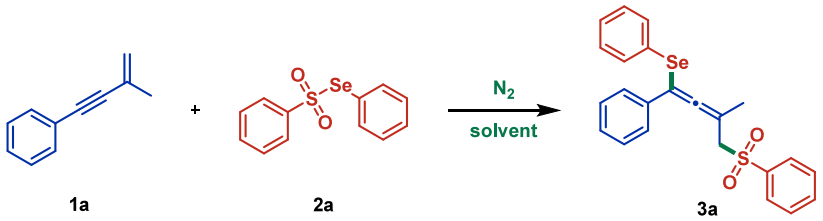
2. RESULTS AND DISCUSSION

Initially, as envisioned, we stirred 1,3-enyne (**1a**) with selenosulfonate (**2a**) in DMSO solvent at room temperature under nitrogen protection using copper catalysis for 12 h. However, the target product (**3a**) was isolated in only 12% yield. (Table 1, entry 1). Encouraged by this promising result, we performed a series of conditional screens. When we added K_3PO_4 to the system, we found that the reaction was completely inhibited (Table 1, entry 2). Next, we screened the solvent types for the reaction and we found that both methanol and water could be used as the optimal solvents with an elevated yield of 27% (Table 1, entries 3−9). Considering that temperature may have a facilitating effect on the reaction, we explored the temperature of the reaction and found that the best yield was obtained when the temperature was increased to 60 °C (Table 1, entries 10−12). On this basis, the yield was further enhanced when we increased the amount of selenosulfonate to 2 equiv (Table 1, entry 13). We tried to add no ligand to the system and found that the yield decreased. However, when both copper and ligand were not added, the yield could still reach 50%, while reducing the amount of selenosulfonate, the yield decreased slightly (Table 1, entries 14−16). Since both methanol and water were the optimal solvents, we envisioned whether a mixture of solvents could be used to further enhance the yield of this reaction. By adjusting the ratio of the two solvents, we found that the yield could be increased to 76% when the ratio of water to methanol was 9:1 (Table 1, entries 17−19).

With the optimized reaction conditions in hand, we explored the substrate scope of 1,3-enynes (Scheme 2). It was found that the substitution position of the methyl group on the benzene ring can have a significant effect on the yields. When the methyl group in the aromatic ring is located at the *meta*- and *para*-positions, target products **3c** and **3d** were obtained in moderate yields. When the aromatic ring with the methyl group substituted at the *ortho*-position, the yield of target product **3b** is only 16% for the hindrance effect. The reactions of *para*-ethyl, *para*-*tert*-butyl, and *meta*-methoxy substituted substrates underwent smoothly to afford corresponding products **3e**−**3g** in moderate yields. It should be noted that the reaction of biphenyl-based 1,3-enynes led to corresponding product **3h** in 73% yield. In addition, pyridines, thiophenes, and halogen-substituted 1,3-enynes with heteroatoms all could be applied to the reaction with **2a** to yield desired products **3i**−**3l** in moderate yields. The reaction of substrate bearing trimethylsilyl group with **2a** could also afford corresponding product **3m** in 30% yield. Unfortunately, the reaction failed when the substrate bearing ester group and carboxylic acid as well as cyclopropane.

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J. Org. Chem. 2024, 89, 8804−8814

Table 1. Optimization of the Reaction Conditions[a]



1a 2a 3a

entry	solvent	catalyst	L	temperature (°C)	yield (%)
1	DMSO	CuI	dtbbpy	rt	12
2	DMSO	CuI	dtbbpy	rt	NR[b]
3	DMF	CuI	dtbbpy	rt	16
4	DCE	CuI	dtbbpy	rt	14
5	CH_3CN	CuI	dtbbpy	rt	12
6	MeOH	CuI	dtbbpy	rt	27
7	H_2O	CuI	dtbbpy	rt	27
8	THF	CuI	dtbbpy	rt	10
9	1,4-dioxane	CuI	dtbbpy	rt	26
10	H_2O	CuI	dtbbpy	40	38
11	H_2O	CuI	dtbbpy	60	40
12	H_2O	CuI	dtbbpy	80	18
13	H_2O	CuI	dtbbpy	60	49[c]
14	H_2O	CuI		60	32[c]
15	H_2O			60	50[c]
16	H_2O			60	46[d]
17	H_2O/CH_3OH 1:1			60	65[c]
18	H_2O/CH_3OH 3:1			60	73[c]
19	H_2O/CH_3OH 9:1			60	76[c]

[a]Reaction conditions: **1a** (0.20 mmol), **2a** (0.40 mmol), H_2O/CH_3OH = 9:1 (2 mL), 60 °C and 12 h. [b]K_3PO_4 is added to the reaction system. [c]**2a** (2 equiv). [d]**2a** (1.5 equiv). dtbbpy ($Ir[FCF_3(CF_3)ppy]_2(dtbbpy)PF_6$).

Next, we investigated the substrate generalization of selenosulfonates (Scheme 3). It was found that when alkyl-substituted selenosulfonates were used, corresponding products **4a**−**4b** could not be obtained. Using aryl-substituted selenosulfonates, corresponding products **4c** and **4d** could be obtained in good yields, for example, when selenosulfonates containing methyl substitution at different positions were used for the reaction, although the position of the substituent would have an effect on the yields, with the adjacent substituted. The reaction is better with adjacent substituted methyl groups than with *para*-substituted methyl groups. When the reaction was carried out using selenosulfonates containing the electron-donating group methoxy substitution, corresponding product **4e** could be obtained in 59% yield. when the reaction was carried out using halogen-substituted selenosulfonates containing electron-absorbing groups, corresponding products **4f** and **4g** were obtained in 65% and 71% yields, respectively. In addition, the selenosulfonates containing naphthalene rings also gave corresponding product **4h** in 70% yield. However, the pyridine selenosulfonate containing heteroatoms could not allow the reaction to take place.

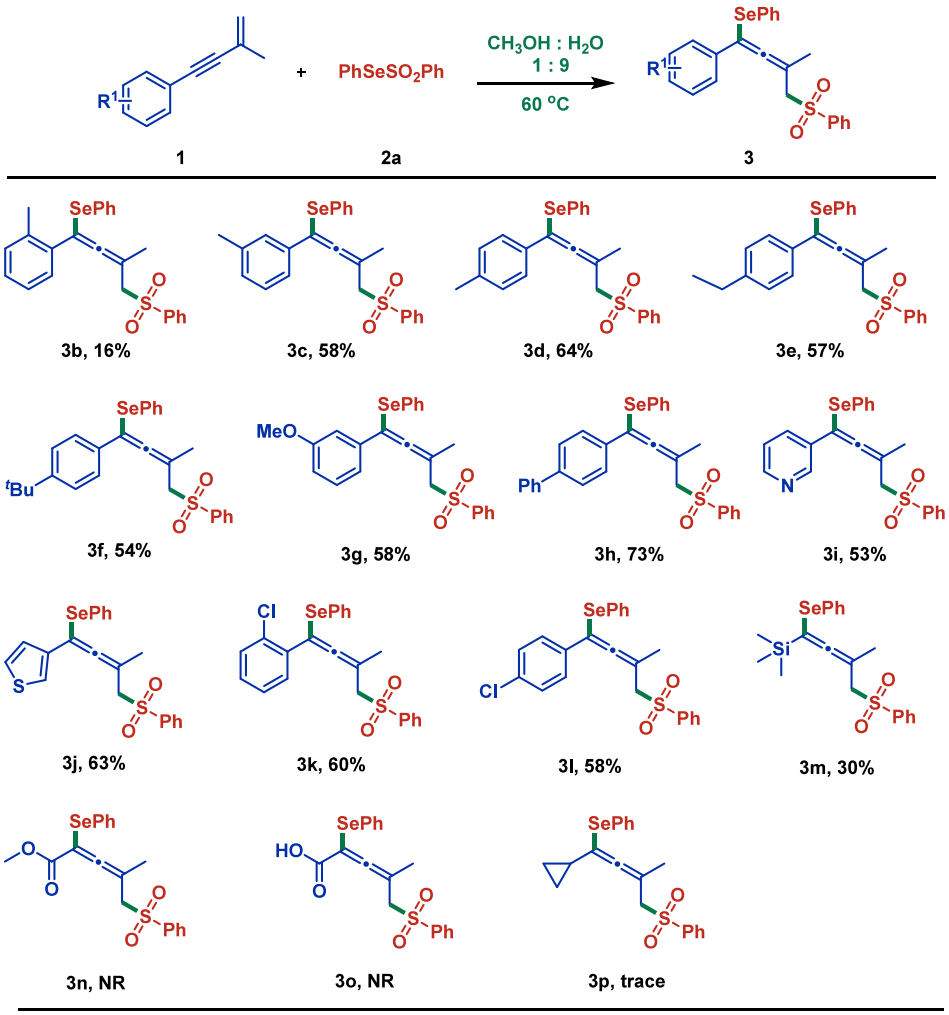
To our surprise, in the process of substrate expansion, pyridine-substituted 1,3-enynes afforded another product, **5a**, in 15% yield (Scheme 4), in addition to the production of allene product **3i**. On the basis of the template reaction conditions, we then carried out optimization screening of the reaction conditions (see SI Table S1). Finally, corresponding product **5a** could be obtained in 68% yield when H_2O was used as the solvent and the temperature was 80 °C.

Subsequently, under optimal conditions, we investigated the functional group tolerance of the reaction (Scheme 5). We found that for methyl with electron-donating effect corresponding products **5b** is obtained in lower yields. Subsequently, the aryl ring of selenosulfonates with electron-donating and electron-absorbing groups on the aromatic ring, respectively, we found that the substrate with methoxy gives target product **5c** in only trace yield, while the halogen-substituted and trifluoromethyl-substituted aryl selenosulfonates give target products **5d**−**5f** in moderate yields. It is possible that the alkyl-substituted trifluoromethyl selenosulfonate could not react smoothly due to a large site barrier. In addition, the naphthalene ring-substituted selenosulfonate also could not allow the reaction to proceed smoothly.

To investigate the feasibility of the reaction, we expanded the reaction scale to 4 and 10 mmol and found that the yield of the target product was 76% and 62%, respectively (Scheme 6).Then a series of controlled experiments were performed to investigate the reaction mechanism. The reaction was completely inhibited by adding TEMPO to the reaction system under standard reaction conditions. The yield of **3a** was dramatically reduced to 15% by adding BHT (Scheme 6 a,b). By replacing the selenosulfonate with a thiosulfonate or with combination of thiosulfonate and diselenide, the reaction still did not occur (Scheme 6 c,d). The pyridine-substituted 1,3-enyne still did not react with the thiosulfonate, but the allene product could be obtained with the thiosulfonate and diselenide (Scheme 6 e,f). The reaction of **3i** with **2a** could lead to allene disulfonated product **5a** in 58% yield (Scheme 6 g).

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J. Org. Chem. 2024, 89, 8804−8814

Scheme 2. Substrate Scope of 1,3-Enynes[a,b]



[a]Reaction conditions: 1a (0.20 mmol), 2a (0.40 mmol), H_2O/CH_3OH = 9:1 (2 mL), 60 °C and 12 h. [b]Isolated yield.

Based on the relevant literatures and the controlled experiments we performed, we propose a possible reaction mechanism (Scheme 7). First, the selenosulfonate undergoes homolytic cleavage under heating conditions to produce the sulfone radical and the selenium radical. Next, the sulfone radical attacks the olefin of 1,3-enynes to give the allyl radical intermediate **A**. Allyl radical intermediate **A** undergoes isomerization to give allene radical intermediate **B**, followed by the capture of intermediate **B** by the selenium radical to give product **3**. When the substrate is a pyridine-substituted 1,3-enynes, the product is further isomerized to give intermediate **C**. Then, the sulfone radical attacks the olefin to generate intermediate **D**. Subsequently, **D** is trapped by the solvent to give intermediate **E**. Following the retro-Micheal reaction of **E**, **5** is observed.

3. CONCLUSION

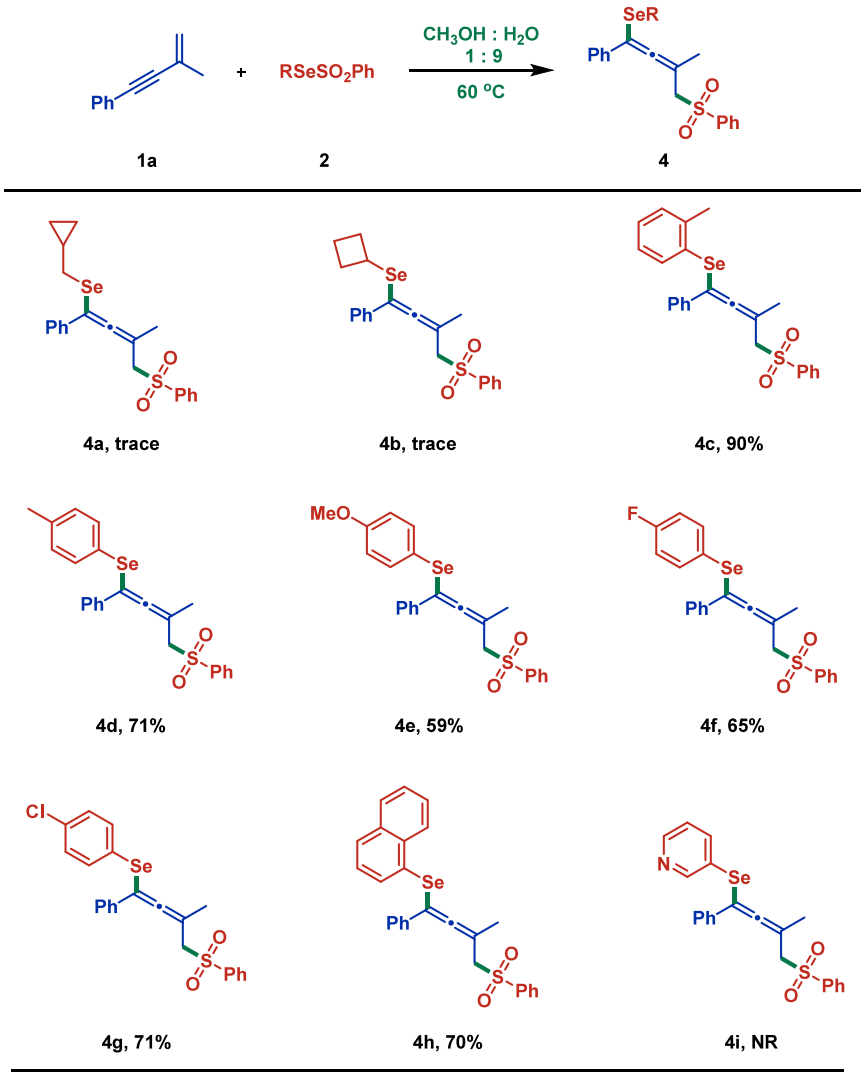
In conclusion, we developed a metal-free/catalyst-free regioselective 1,3/1,4-bifunctionalization reaction of 1,3-enynes with selenosulfonate. By modulating the substrate, both 1,4 bifunctionalization of 1,3-enynes and 1,3-bifunctionalization of 1,3-enynes can be achieved to construct a allene compound and a disulfonyl conjugated diene. The reaction does not require metal catalysis, is simple to operate, uses a green solvent, and provides a new idea for the construction of allene compounds and conjugated dienes.

4. EXPERIMENTAL SECTION

4.1. General Experimental Information. Unless otherwise noted, all commercially available compounds were used as provided without further purification. Solvents for chromatography were analytical grade and used without further purification. All the solvents for routine isolation of products and chromatography were reagent grade. Analytical thin-layer chromatography (TLC) was performed on silica gel, irradiation with UV light. For column chromatography, 300−400 mesh silica gel was used. ^{1}H NMR and $^{13}C\{^1H\}$ NMR spectra were recorded on a BRUKER 400 MHz spectrometer and a BRUKER 300 MHz spectrometer in $CDCl_3$. Chemical shifts (δ) were reported referenced to an internal tetramethylsilane standard or the $CDCl_3$ residual peak (δ 7.26) for ^{1}H NMR. Chemical shifts of $^{13}C\{^1H\}$ NMR are reported relative to $CDCl_3$ (δ 77.16). Data are reported in the following order: chemical shift (δ) in ppm; multiplicities are indicated s (singlet), bs (broad singlet), d (doublet), t (triplet), m (multiplet); coupling constants (J) are in Hertz (Hz). Melting points were measured on an electrothermal digital melting point apparatus and were uncorrected. IR spectra were recorded on a BRUKER VERTEX 70 spectrophotometer and are reported in terms of frequency of absorption (cm^{-1}). HRMS spectra were obtained by using GCT Premier TOF-MS with CI source or BRUKER microTOF-Q III instrument with ESI source. Single-crystal X-ray

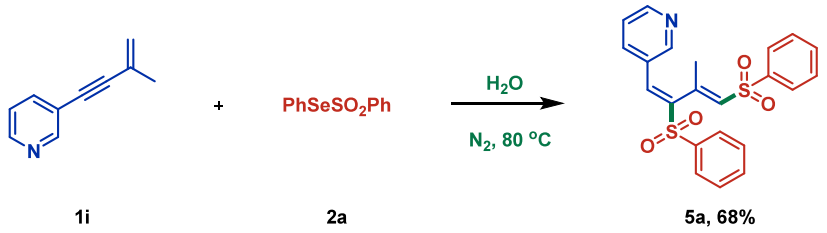
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Scheme 3. Substrate Scope of Selenosulfonates[a,b]



[a]Reaction conditions: 1a (0.20 mmol), 2 (0.40 mmol), H_2O/CH_3OH = 9:1 (2 mL), 60 °C and 12 h. [b]Isolated yield.

Scheme 4. 1,3-Difunctionalization to Construct Bis-sulfonyl Conjugated Dienes[a]



[a]Reaction conditions: **1i** (0.20 mmol), **2a** (0.40 mmol), H_2O (2 mL), 80 °C, and 12 h.

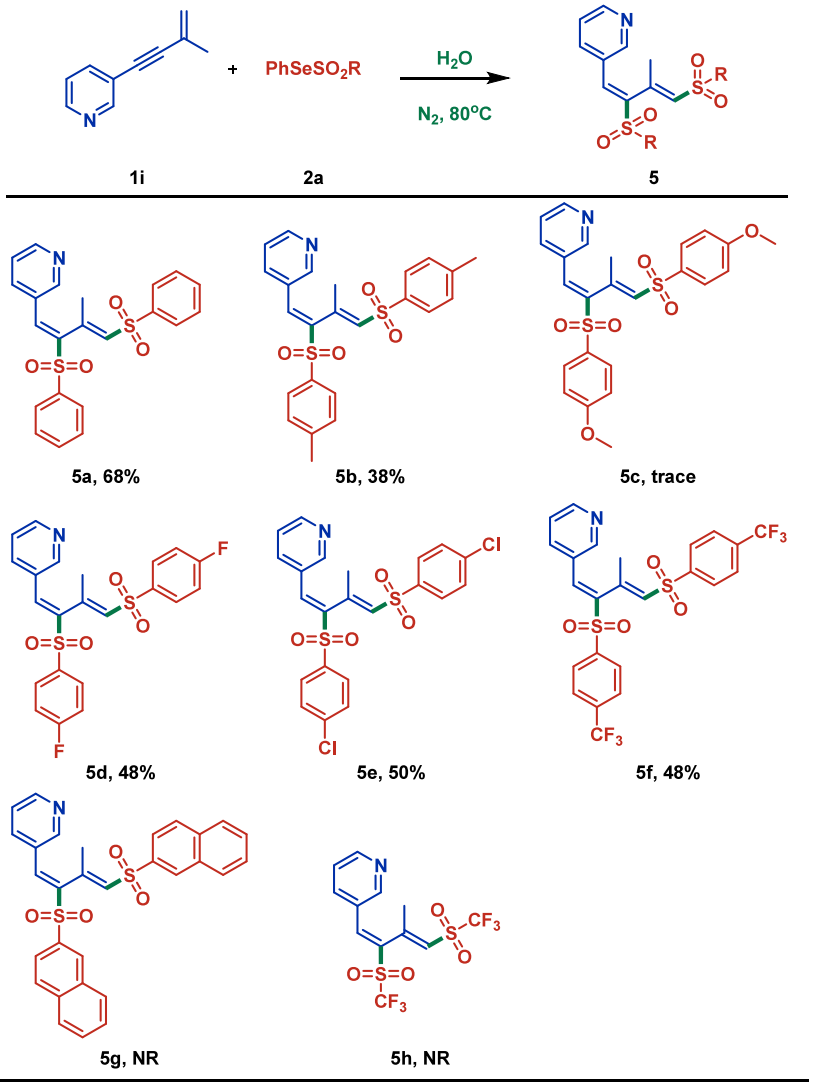
diffraction data for all crystals were mounted on Bruker Smart CCD diffractometer with a graphite monochromated Mo Kα radiation (λ = 0.71073 Å).

4.2. General Procedure for the Synthesis of 3 and 5. In a nitrogen-filled glovebox, benzene ring-substituted 1,3-enynes (0.2 mmol) and selenosulfonate (0.4 mmol) were added to a screw-capped 8 mL reaction flask, followed by methanol (0.2 mL) and water (1.8 mL), and the mixture was allowed to stir for 12 h at 60 °C. At the end of the reaction, it was diluted with ethyl acetate (10.0 mL). The organic phase was washed with water (5.0 mL mL × 3). The organic layer was dried over Na_2SO_4, filtered and concentrated. The residue was purified by fast chromatography (ethyl acetate/petroleum ether (V/V = 1/5) to give the target product (0.0669 g) in 76% yield.

In a nitrogen-filled glovebox, pyridine-substituted 1,3-enynes (0.2 mmol), selenosulfonates (0.4 mmol), and then water (2 mL) were added to a screw-capped 8 mL reaction flask, and the mixture was allowed to stir for 12 h at 80 °C. At the end of the reaction, it was diluted with ethyl acetate (10.0 mL). The organic phase was washed with water (5.0 mL × 3). The organic layer was dried over Na_2SO_4, filtered and concentrated. The residue was purified by fast chromatography (ethyl acetate/petroleum ether (V/V = 1/2) to give the target product (0.0578 g) in 68% yield.

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J. Org. Chem. 2024, 89, 8804−8814

Scheme 5. Substrate Scope of Pyridine-Substituted 1,3-Enynes[a,b]



[a]Reaction conditions: **1i** (0.20 mmol), **2a** (0.40 mmol), H_2O (2 mL), 80 °C, and 12 h. [b]Isolated yield.

*4.2.1. (S)-(3-Methyl-1-phenyl-4-(phenylsulfonyl)buta-1,2-dien-1-yl)(phenyl)selane (**3a**).* Yellow oil, 76% yield (0.0669 g, 0.2 mmol scale). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.87−7.82 (m, 2H), 7.53 (d, J = 6.4 Hz, 1H), 7.45−7.40 (m, 4H), 7.30−7.27 (m, 2H), 7.24 (s, 6H), 3.65 (d, J = 13.9 Hz, 1H), 3.51 (d, J = 14.0 Hz, 1H), 1.80 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 204.3, 138.5, 134.9, 134.4, 134.2, 133.7, 130.1, 129.2, 129.1, 128.4, 128.2, 127.9, 127.7, 94.0, 60.8, 18.4. ^{77}Se NMR (76 MHz, $CDCl_3$) δ 405.52. IR (neat): ν = 3055, 2981, 1734, 1306, 1237, 1143, 741, 686, 527 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{23}H_{20}O_2SSe$ [M]: 440.0349, found: 440.0349.

*4.2.2. (S)-(3-Methyl-4-(phenylsulfonyl)-1-(o-tolyl)buta-1,2-dien-1-yl)(phenyl)selane (**3b**).* Yellow oil, 16% yield (0.0145 g, 0.2 mmol scale). ^{1}H NMR (400 MHz, DMSO-d_6) δ 7.73 (d, J = 7.7 Hz, 2H), 7.47−7.44 (m, 1H), 7.41 (m, 2H), 7.38−7.28 (m, 5H), 7.13 (m, 4H), 3.96−3.83 (m, 2H), 2.17 (s, 3H), 1.67 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, DMSO-d_6) δ 203.2, 138.7, 137.2, 135.9, 134.5, 134.0, 130.8, 130.2, 130.0, 129.7, 129.4, 128.6, 128.3, 128.0, 126.3, 95.0, 92.2, 59.6, 20.5, 18.7. ^{77}Se NMR (76 MHz, DMSO-d_6) δ 456.47. IR (neat): ν = 3059, 2919, 1731, 1308, 1145, 738, 686, 525 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{24}H_{22}O_2SSe$ [M]: 454.0506, found: 454.0506.

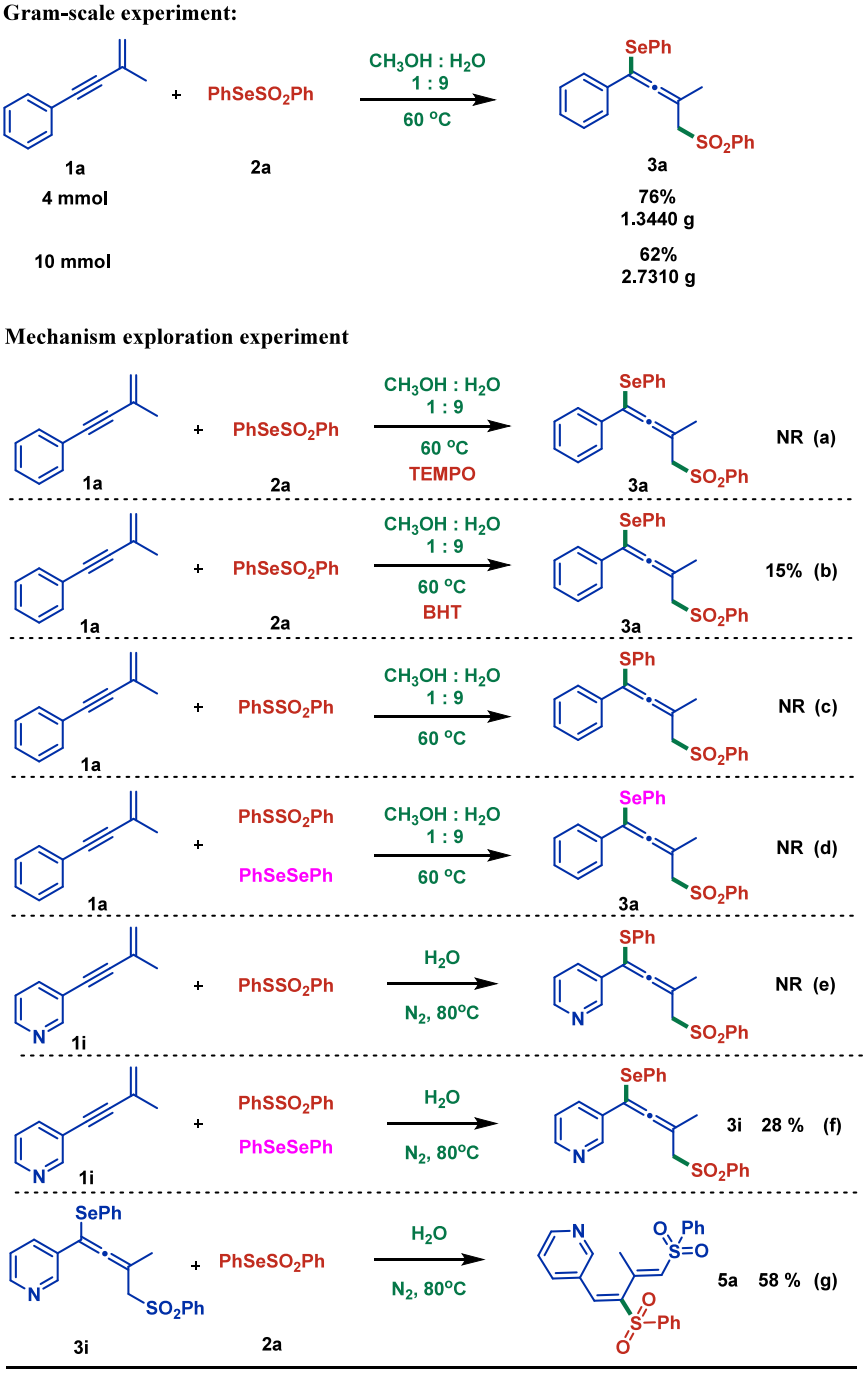
*4.2.3. (S)-(3-Methyl-4-(phenylsulfonyl)-1-(m-tolyl)buta-1,2-dien-1-yl)(phenyl)selane (**3c**).* Yellow oil, 58% yield (0.0526 g, 0.2 mmol scale). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.76−7.70 (m, 2H), 7.40 (t, J = 7.5 Hz, 1H), 7.36−7.27 (m, 4H), 7.20−7.13 (m, 3H), 7.10 (s, 1H), 7.03 (t, J = 7.6 Hz, 1H), 6.93 (t, J = 8.2 Hz, 2H), 3.53 (d, J = 13.9 Hz, 1H), 3.39 (d, J = 14.0 Hz, 1H), 2.23 (s, 3H), 1.67 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 204.2, 138.6, 138.1, 134.4, 134.2, 133.6, 130.3, 129.11, 129.09, 128.8, 128.4, 128.3, 128.2, 127.9, 124.9, 93.8, 60.8, 21.5, 18.5. ^{77}Se NMR (76 MHz, $CDCl_3$) δ 406.00. IR (neat): ν = 2980, 2920, 1733, 1304, 1143, 740, 687 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{24}H_{22}O_2SSe$ [M]: 454.0506, found: 454.0506.

*4.2.4. (S)-(3-Methyl-4-(phenylsulfonyl)-1-(p-tolyl)buta-1,2-dien-1-yl)(phenyl)selane (**3d**).* Yellow oil, 64% yield (0.0581 g, 0.2 mmol scale). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.86 (d, J = 9.5 Hz, 2H), 7.60−7.49 (m, 2H), 7.43 (m, 4H), 7.31−7.27 (m, 2H), 7.15 (d, J = 8.2 Hz, 2H), 7.06 (d, J = 8.0 Hz, 2H), 3.67 (d, J = 13.9 Hz, 1H), 3.53 (d, J = 13.9 Hz, 1H), 2.35 (s, 3H), 1.80 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 204.2, 138.6, 137.9, 134.0, 133.7, 131.4, 130.3, 129.2, 129.2, 129.1, 128.3, 127.8, 127.6, 98.8, 93.7, 60.9, 21.2, 18.5. ^{77}Se NMR (76 MHz, $CDCl_3$) δ 404.19. IR (neat): ν = 2981, 2920, 1733, 1303, 1143, 1081, 740, 686, 580 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{24}H_{22}O_2SSe$ [M]: 454.0506, found: 454.0506.

*4.2.5. (S)-(1-(4-Ethylphenyl)-3-methyl-4-(phenylsulfonyl)buta-1,2-dien-1-yl)(phenyl)selane (**3e**).* Yellow oil, 57% yield (0.0534 g, 0.2 mmol scale). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.88−7.83 (m, 2H), 7.58−7.50 (m, 2H), 7.43 (d, J = 7.8 Hz, 4H), 7.31−7.27 (m, 2H), 7.18 (d, J = 8.2 Hz, 2H), 7.09 (d, J = 8.2 Hz, 2H), 3.66 (d, J = 13.9

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J. Org. Chem. 2024, 89, 8804−8814

Scheme 6. Mechanism Verification and Control Experiments



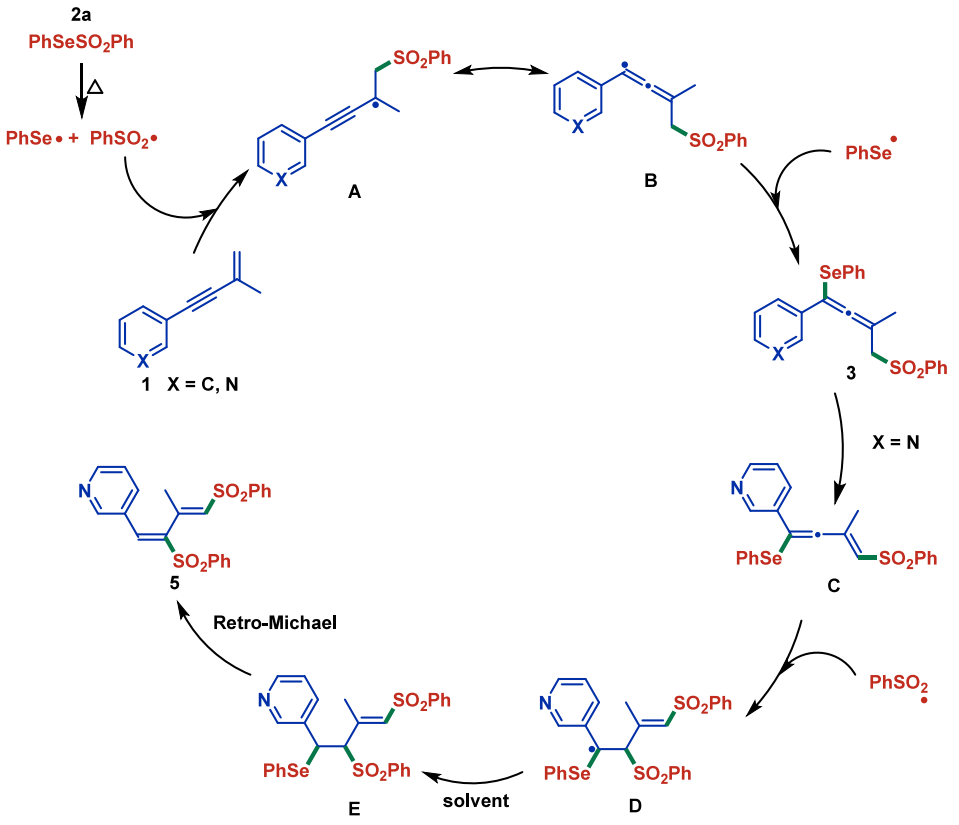
Hz, 1H), 3.52 (d, J = 14.0 Hz, 1H), 2.65 (q, J = 7.6 Hz, 2H), 1.79 (s, 3H), 1.25 (t, J = 7.6 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 204.2, 144.3, 138.6, 134.0, 133.6, 131.7, 129.2, 129.1, 128.2, 128.0, 127.8, 127.6, 98.9, 93.7, 60.9, 28.6, 18.5, 15.6. ^{77}Se NMR (76 MHz, $CDCl_3$) δ 404.45. IR (neat): ν = 2965, 1733, 1307, 1239, 1145, 738, 687, 526 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{25}H_{24}O_2SSe$ [M]: 468.0662, found: 468.0662.

*4.2.6. (S)-(1-(4-(tert-Butyl)phenyl)-3-methyl-4-(phenylsulfonyl)-buta-1,2-dien-1-yl)(phenyl)selane (**3f**).* Yellow oil, 54% yield (0.0536 g, 0.2 mmol scale). ^1H NMR (400 MHz, $CDCl_3$) δ 7.83 (d, J = 7.2 Hz, 2H), 7.51 (t, J = 7.9 Hz, 3H), 7.42−7.36 (m, 6H), 7.28 (s, 1H), 7.24 (s, 1H), 7.22 (s, 1H), 3.63 (d, J = 13.9 Hz, 1H), 3.50 (d, J = 13.9 Hz, 1H), 1.76 (s, 3H), 1.32 (s, 9H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 204.4, 151.1, 137.3, 134.1, 129.6, 129.2, 129.1, 128.8, 128.2, 127.8, 127.4, 127.0, 125.4, 93.7, 60.9, 34.6, 31.3, 18.5. ^{77}Se NMR (76 MHz, $CDCl_3$) δ 403.82. IR (neat): ν = 2962, 2905, 1307, 1239, 1142, 1080, 735, 686, 528 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{27}H_{28}O_2SSe$ [M]: 496.0975, found: 496.0975.

*4.2.7. (S)-(1-(3-Methoxyphenyl)-3-methyl-4-(phenylsulfonyl)-buta-1,2-dien-1-yl)(phenyl)selane (**3g**).* Yellow oil, 58% yield (0.0545 g, 0.2 mmol scale). ^1H NMR (400 MHz, $CDCl_3$) δ 7.71 (d, J = 7.5 Hz, 2H), 7.39 (t, J = 6.8 Hz, 1H), 7.33−7.27 (m, 4H), 7.13 (q, J = 6.8, 6.1 Hz, 3H), 7.03 (t, J = 8.2 Hz, 1H), 6.92 (s, 1H), 6.76 (d, J = 7.8 Hz, 1H), 6.66 (d, J = 11.0 Hz, 1H), 3.68 (s, 3H), 3.52 (d, J = 13.8 Hz, 1H), 3.38 (d, J = 13.8 Hz, 1H), 1.63 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 204.6, 159.7, 138.7, 135.9, 134.1, 133.7, 130.2, 129.4, 129.2, 129.1, 128.1, 127.9, 120.2, 113.9, 113.3, 93.7, 60.8, 55.4, 18.6. ^{77}Se NMR (76 MHz, $CDCl_3$) δ 406.43. IR (neat): ν = 3060, 2980, 1733, 1238, 1144, 1042, 739, 687 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{24}H_{22}O_3SSe$ [M]: 470.0455, found: 470.0455.

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J. Org. Chem. 2024, 89, 8804−8814

Scheme 7. Proposed Reaction Mechanism



*4.2.8. (S)-(1-([1,1′-Biphenyl]-4-yl)-3-methyl-4-(phenylsulfonyl)-buta-1,2-dien-1-yl)(phenyl)selane (**3h**).* Yellow oil, 73% yield (0.0754 g, 0.2 mmol scale). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.78−7.73 (m, 2H), 7.51−7.46 (m, 2H), 7.41−7.30 (m, 9H), 7.25−7.13 (m, 6H), 3.57 (d, J = 13.9 Hz, 1H), 3.42 (d, J = 13.9 Hz, 1H), 1.71 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 204.7, 140.8, 140.5, 138.6, 134.1, 133.7, 133.4, 129.24, 129.17, 128.9, 128.3, 128.1, 128.0, 127.6, 127.1, 127.0, 94.0, 60.8, 18.5. ^{77}Se NMR (76 MHz, $CDCl_3$) δ 405.23. IR (neat): ν = 2981, 1732, 1304, 1142, 1080, 744, 686, 603 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{29}H_{24}O_2SSe$ [M]: 516.0662, found: 516.0662.

*4.2.9. (S)-3-(3-Methyl-1-(phenylselanyl)-4-(phenylsulfonyl)buta-1,2-dien-1-yl)pyridine (**3i**).* Yellow oil, 53% yield (0.0468 g, 0.2 mmol scale). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.45 (d, J = 6.2 Hz, 2H), 7.83 (d, J = 8.5 Hz, 2H), 7.62 (d, J = 8.0 Hz, 1H), 7.51 (d, J = 6.3 Hz, 1H), 7.45−7.39 (m, 4H), 7.31−7.26 (m, 2H), 7.19 (m, 1H), 3.66 (d, J = 13.9 Hz, 1H), 3.54 (d, J = 13.9 Hz, 1H), 1.82 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 204.9, 148.45, 148.36, 138.3, 138.1, 135.3, 134.1, 133.8, 129.3, 129.2, 128.3, 128.2, 123.3, 94.6, 60.5, 18.5. ^{77}Se NMR (76 MHz, $CDCl_3$) δ 410.15. IR (neat): ν = 2980, 2921, 1302, 1141, 1080, 685, 523 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{22}H_{19}NO_2SSe$ [M]: 441.0302, found: 441.0302.

*4.2.10. (S)-3-(3-Methyl-1-(phenylselanyl)-4-(phenylsulfonyl)-buta-1,2-dien-1-yl)thiophene (**3j**).* Yellow oil, 63% yield (0.0561 g, 0.2 mmol scale). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.86 (d, J = 7.2 Hz, 2H), 7.70−7.66 (m, 1H), 7.47−7.40 (m, 5H), 7.25−7.22 (m, 2H), 7.19 (m, 2H), 6.85 (d, J = 6.3 Hz, 1H), 3.66 (d, J = 13.8 Hz, 1H), 3.53 (d, J = 13.8 Hz, 1H), 1.75 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 204.6, 138.8, 136.1, 135.0, 133.9, 130.4, 129.3, 129.2, 128.2, 127.9, 127.1, 126.0, 123.6, 93.4, 93.1, 60.9, 18.7. ^{77}Se NMR (76 MHz, $CDCl_3$) δ 417.61. IR (neat): ν = 2979, 2920, 1731, 1307, 1240, 1145, 1080, 736, 685, 526 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{21}H_{18}O_2S_2Se$ [M]: 445.9913, found: 445.9913.

*4.2.11. (S)-(1-(2-Chlorophenyl)-3-methyl-4-(phenylsulfonyl)buta-1,2-dien-1-yl)(phenyl)selane (**3k**).* Yellow oil, 60% yield (0.0568 g, 0.2 mmol scale). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.80−7.73 (m, 2H), 7.47 (m, 2H), 7.39−7.30 (m, 4H), 7.28 (m, 2H), 7.23 (m, 2H), 7.21−7.16 (m, 2H), 3.63 (d, J = 14.0 Hz, 1H), 3.57 (d, J = 14.0 Hz, 1H), 1.87 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 204.1, 137.5, 134.6, 133.9, 133.3, 132.8, 130.8, 129.9, 129.8, 129.0, 128.9, 128.6, 128.2, 128.1, 126.6, 93.6, 92.8, 60.5, 17.8. ^{77}Se NMR (76 MHz, $CDCl_3$) δ 470.10. IR (neat): ν = 2981, 2917, 1733, 1310, 1239, 1146, 739, 687, 526 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{23}H_{19}ClO_2SSe$ [M]: 473.9960, found: 473.9960.

*4.2.12. (S)-(1-(4-Chlorophenyl)-3-methyl-4-(phenylsulfonyl)buta-1,2-dien-1-yl)(phenyl)selane (**3l**).* Yellow oil, 58% yield (0.0550 g, 0.2 mmol scale). ^{1}H NMR (400 MHz, DMSO-d_6) δ 7.04−6.98 (m, 2H), 6.77 (t, J = 7.4 Hz, 1H), 6.66 (t, J = 7.7 Hz, 2H), 6.51 (m, 2H), 6.40 (m, 5H), 6.34−6.30 (m, 2H), 3.19 (d, J = 2.2 Hz, 2H), 0.81 (s, 3H). ^{13}C{^{1}H} NMR (101 MHz, DMSO-d_6) δ 205.8, 139.1, 134.4, 133.6, 133.1, 133.0, 130.0, 129.9, 129.8, 129.4, 129.0, 128.3, 128.2, 95.8, 94.2, 59.3, 18.9. ^{77}Se NMR (76 MHz, DMSO-d_6) δ 402.21. IR (neat): ν = 2981, 2916, 1732, 1309, 1240, 1146, 1016, 739, 687, 527 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{23}H_{19}ClO_2SSe$ [M]: 473.9960, found: 473.9960.

*4.2.13. (S)-Trimethyl(3-methyl-1-(phenylselanyl)-4-(phenylsulfonyl)buta-1,2-dien-1-yl)silane (**3m**).* Yellow oil, 30% yield (0.0262 g, 0.2 mmol scale). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.91−7.86 (m, 2H), 7.67−7.60 (m, 1H), 7.58−7.51 (m, 2H), 7.47−7.42 (m, 2H), 7.33−7.26 (m, 3H), 3.50 (d, J = 13.9 Hz, 1H), 3.39 (d, J = 13.9 Hz, 1H), 1.61 (s, 3H), 0.11 (s, 9H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 205.2, 139.4, 135.0, 133.7, 129.9, 129.3, 128.8, 128.2, 127.9, 89.6, 87.6, 60.9, 17.7. ^{77}Se NMR (76 MHz, $CDCl_3$) δ 388.90. IR (neat): ν = 2956, 1310, 1245, 1147, 840, 737, 687, 569, 524 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{20}H_{24}O_2SSeSi$ [M]: 436.0431, found: 436.0431.

*4.2.14. (S)-(3-Methyl-1-phenyl-4-(phenylsulfonyl)buta-1,2-dien-1-yl)(o-tolyl)selane (**4c**).* Yellow oil, 90% yield (0.0818 g, 0.2 mmol scale). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.82 (m, 2H), 7.51 (d, J = 7.5 Hz, 1H), 7.41−7.33 (m, 3H), 7.26−7.20 (m, 7H), 7.05 (m, 1H), 3.56 (d, J = 13.9 Hz, 1H), 3.42 (d, J = 14.0 Hz, 1H), 2.41 (s, 3H), 1.75 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 203.7, 140.5, 138.4, 134.9,

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J. Org. Chem. 2024, 89, 8804−8814

134.4, 133.6, 131.2, 130.1, 129.2, 128.5, 128.2, 128.0, 127.6, 126.6, 98.4, 94.0, 60.6, 22.6, 18.2. ^{77}Se NMR (76 MHz, $CDCl_3$) δ 359.68. IR (neat): ν = 3057, 2920, 1734, 1443, 1308, 1145, 742, 684, 523 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{24}H_{22}O_2SSe$ [M]: 454.0506, found: 454.0506.

*4.2.15. (S)-(3-Methyl-1-phenyl-4-(phenylsulfonyl)buta-1,2-dien-1-yl)(p-tolyl)selane (**4d**).* Yellow oil, 71% yield (0.0645 g, 0.2 mmol scale). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.87−7.81 (m, 2H), 7.53 (t, J = 8.4 Hz, 1H), 7.40 (t, J = 7.8 Hz, 2H), 7.32 (d, J = 8.2 Hz, 2H), 7.21 (d, J = 3.0 Hz, 5H), 7.07 (d, J = 7.8 Hz, 2H), 3.63 (d, J = 13.9 Hz, 1H), 3.47 (d, J = 14.0 Hz, 1H), 2.34 (s, 3H), 1.77 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 203.8, 138.4, 138.3, 134.7, 134.5, 133.7, 129.9, 129.2, 128.4, 128.2, 127.9, 127.6, 126.2, 94.3, 60.8, 21.2, 18.4. ^{77}Se NMR (76 MHz, $CDCl_3$) δ 396.30. IR (neat): ν = 2975, 2919, 1303, 1144, 1080, 688, 603, 526, 486 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{24}H_{22}O_2SSe$ [M]: 454.0506, found: 454.0506.

*4.2.16. (S)-(4-Methoxyphenyl)(3-methyl-1-phenyl-4-(phenylsulfonyl)buta-1,2-dien-1-yl)selane (**4e**).* Yellow oil, 59% yield (0.0555 g, 0.2 mmol scale). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.74 (m, 2H), 7.48−7.42 (m, 1H), 7.35−7.28 (m, 4H), 7.13 (m, 3H), 7.10−7.06 (m, 2H), 6.75−6.68 (m, 2H), 3.71 (s, 3H), 3.50 (d, J = 13.9 Hz, 1H), 3.35 (d, J = 13.9 Hz, 1H), 1.63 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 202.7, 160.2, 138.4, 137.3, 134.5, 133.7, 129.2, 128.4, 128.2, 127.9, 127.5, 119.8, 114.7, 100.7, 95.0, 60.9, 55.4, 18.4. ^{77}Se NMR (76 MHz, $CDCl_3$) δ 388.55. IR (neat): ν = 2972, 2934, 1732, 1585, 1488, 1243, 1144, 1082,686, 519 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{24}H_{22}O_3SSe$ [M]: 470.0455, found: 470.0455.

*4.2.17. (S)-(4-Fluorophenyl)(3-methyl-1-phenyl-4-(phenylsulfonyl)buta-1,2-dien-1-yl)selane (**4f**).* Yellow oil, 65% yield (0.0596 g, 0.2 mmol scale). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.84 (m, 2H), 7.54 (t, J = 7.5 Hz, 1H), 7.45−7.38 (m, 4H), 7.23 (s, 5H), 6.99−6.91 (m, 2H), 3.66 (d, J = 13.8 Hz, 1H), 3.55 (d, J = 13.9 Hz, 1H), 1.77 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 203.8, 162.9 (d, J = 253.0 Hz), 138.6, 136.7 (d, J = 8.0 Hz), 134.2, 133.8, 129.2, 128.5, 128.2, 128.0, 127.6, 124.5 (d, J = 3.0 Hz), 116.3 (d, J = 21.0 Hz), 94.3, 60.8, 18.6. ^{77}Se NMR (76 MHz, $CDCl_3$) δ 397.56. ^{19}F NMR (376 MHz, $CDCl_3$) δ −112.73. IR (neat): ν = 2982, 1732, 1484, 1306, 1224, 1147, 1043, 689, 507 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{23}H_{19}FO_2SSe$ [M]: 458.0255, found: 458.0255.

*4.2.18. (S)-(4-Chlorophenyl)(3-methyl-1-phenyl-4-(phenylsulfonyl)buta-1,2-dien-1-yl)selane (**4g**).* Yellow oil, 71% yield (0.0673 g, 0.2 mmol scale). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.78 (m, 2H), 7.47 (m, 2H), 7.34 (t, J = 6.9 Hz, 2H), 7.24 (d, J = 2.7 Hz, 1H), 7.19−7.18 (m, 2H), 7.17−7.14 (m, 4H), 7.12 (d, J = 1.9 Hz, 1H), 3.65 (d, J = 13.9 Hz, 1H), 3.55 (d, J = 13.9 Hz, 1H), 1.75 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 205.0, 138.6, 136.0, 134.9, 134.1, 134.0, 133.8, 129.3, 129.2, 128.5, 128.2, 128.1, 127.7, 98.4, 93.6, 60.8, 18.7. ^{77}Se NMR (76 MHz, $CDCl_3$) δ 403.37. IR (neat): ν = 2980, 1732, 1304, 1146, 1084, 689, 604, 486 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{23}H_{19}ClO_2SSe$ [M]: 473.9960, found: 473.9960.

*4.2.19. (S)-(3-Methyl-1-phenyl-4-(phenylsulfonyl)buta-1,2-dien-1-yl)(naphthalen-1-yl)selane (**4h**).* Yellow oil, 70% yield (0.0686 g, 0.2 mmol scale). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.36 (m, 1H), 7.93−7.81 (m, 3H), 7.76−7.70 (m, 2H), 7.58 (m, 2H), 7.53−7.48 (m, 1H), 7.43−7.34 (m, 3H), 7.28 (m, 3H), 7.25−7.21 (m, 2H), 3.01 (d, J = 14.0 Hz, 1H), 2.85 (d, J = 13.9 Hz, 1H), 1.30 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 202.0, 138.3, 135.6, 134.6, 134.5, 134.0, 133.5, 129.9, 129.1, 129.0, 128.7, 128.5, 128.2, 128.1, 128.0, 127.4, 127.1, 126.5, 125.9, 99.9, 95.7, 60.0, 17.8. ^{77}Se NMR (76 MHz, $CDCl_3$) δ 333.73. IR (neat): ν = 3053, 2924, 1732, 1443, 1306, 1145, 1081, 1024, 765, 687, 526 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{27}H_{22}O_2SSe$ [M]: 490.0506, found: 490.0506.

*4.2.20. 3-((1E,3E)-3-Methyl-2,4-bis(phenylsulfonyl)buta-1,3-dien-1-yl)pyridine (**5a**).* Yellow solid, 68% yield (0.0578 g, 0.2 mmol scale). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.59−8.50 (m, 2H), 7.79−7.72 (m, 2H), 7.69−7.63 (m, 2H), 7.54 (m, 5H), 7.49 (t, J = 8.1 Hz, 1H), 7.29 (t, J = 7.9 Hz, 2H), 7.14 (m, 1H), 6.00 (d, J = 1.4 Hz, 1H), 2.07 (d, J = 1.4 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 151.8, 151.6, 144.6, 143.0, 140.6, 137.4, 135.6, 134.8, 134.0, 134.01, 133.96, 129.5, 129.3, 128.6, 127.7, 127.6, 123.8, 18.3. IR (neat): ν = 2956, 1290, 1139, 1081, 781, 665, 533 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{22}H_{19}NO_4S_2$ [M]: 425.0755, found: 425.0755.

*4.2.21. 3-((1E,3E)-3-Methyl-2,4-ditosylbuta-1,3-dien-1-yl)-pyridine (**5b**).* Yellow solid, 38% yield (0.0345 g, 0.2 mmol scale). Mp: 97.0−97.5 °C. ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.64 (d, J = 1.8 Hz, 1H), 8.59 (d, J = 4.7 Hz, 1H), 7.71 (d, J = 8.7 Hz, 3H), 7.65 (d, J = 9.8 Hz, 1H), 7.51 (d, J = 8.3 Hz, 2H), 7.41 (d, J = 8.1 Hz, 2H), 7.22 (m, 1H), 7.15 (d, J = 8.1 Hz, 2H), 6.05 (s, 1H), 2.52 (s, 3H), 2.40 (s, 3H), 2.14 (d, J = 1.3 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 151.8, 151.4, 145.2, 145.0, 144.1, 143.3, 137.7, 135.6, 134.9, 134.3, 133.5, 130.0, 129.9, 128.7, 127.8, 127.6, 123.8, 21.7, 21.7, 18.3. IR (neat): ν = 2959, 2922, 1591, 1410, 1291, 1141, 1082, 779, 663, 534 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{24}H_{23}NO_4S_2$ [M]: 453.1608, found: 453.1608.

*4.2.22. 3-((1E,3E)-2,4-bis((4-fluorophenyl)sulfonyl)-3-methylbuta-1,3-dien-1-yl)pyridine (**5d**).* Yellow solid, 48% yield (0.0443 g, 0.2 mmol scale). Mp: 109.5−110.0 °C. ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.56 (d, J = 4.8 Hz, 2H), 7.83−7.73 (m, 2H), 7.69−7.55 (m, 4H), 7.21 (m, 3H), 7.02 (m, 2H), 6.00 (d, J = 1.4 Hz, 1H), 2.11 (d, J = 1.3 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 166.0 (d, J = 256.0 Hz), 165.9 (d, J = 256.0 Hz), 151.7, 151.6, 144.4, 142.8, 136.6 (d, J = 4.0 Hz), 135.8, 134.8, 134.3, 133.4(d, J = 3.0 Hz), 131.6 (d, J = 10.0 Hz), 130.5(d, J = 9.0 Hz), 127.6, 123.8, 117.0, 116.9 (d, J = 22.0 Hz), 116.8(d, J = 23.0 Hz), 18.3. ^{19}F NMR (376 MHz, $CDCl_3$) δ −101.64, −102.22. IR (neat): ν = 2971, 1904, 1227, 1144, 1077, 837, 596, 534 cm^{-1}. HRMS (EI-TOF) m/z: calcd. for $C_{22}H_{17}F_2NO_4S_2$ [M]: 461.0567, found: 461.0567.

*4.2.23. 3-((1E,3E)-2,4-Bis((4-chlorophenyl)sulfonyl)-3-methylbuta-1,3-dien-1-yl)pyridine (**5e**).* Yellow solid, 50% yield (0.0493 g, 0.2 mmol scale). Mp: 129.5−130.0 °C.^{1}H NMR (400 MHz, chloroform-d) δ 8.65 (s, 2H), 7.75 (d, J = 9.1 Hz, 3H), 7.66 (m, 1H), 7.64−7.54 (m, 4H), 7.39−7.30 (m, 2H), 7.28 (m, 1H), 6.03 (m, 1H), 2.21 (d, J = 1.5 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, chloroform-d) δ 151.8, 144.7, 142.4, 141.0, 138.9, 135.8, 135.7, 134.7, 134.7, 130.0, 129.9, 129.7, 129.0, 18.4. IR (neat): ν = 2981, 2907, 1306, 1148, 1081, 754, 631, 583 cm^{-1}. HRMS (ESI) m/z: calcd. for $C_{22}H_{17}Cl_2NO_4S_2Na$ [M + Na]$^+$: 515.9868, found: 515.9871.

*4.2.24. 3-((1E,3E)-3-Methyl-2,4-bis((4-(trifluoromethyl)phenyl)-sulfonyl)buta-1,3-dien-1-yl)pyridine (**5f**).* Yellow solid, 48% yield (0.0539 g, 0.2 mmol scale). Mp: 106.5−107.0 °C. ^{1}H NMR (400 MHz, chloroform-d) δ 8.82−8.55 (m, 2H), 7.95 (d, J = 8.4 Hz, 2H), 7.88 (d, J = 8.3 Hz, 2H), 7.83−7.78 (m, 3H), 7.69−7.58 (m, 3H), 7.32−7.26 (m, 1H), 6.04 (d, J = 1.5 Hz, 1H), 2.25 (d, J = 1.5 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, chloroform-d) δ 152.0, 151.8, 145.5, 143.8, 141.8, 141.1, 135.9, 135.8, 135.8 (d, J = 33.3 Hz), 135.7 (d, J = 33.3 Hz), 134.3, 129.2, 128.2, 127.3, 126.7 (q, J = 3.6 Hz), 126.4 (q, J = 3.7 Hz), 123.9, 121.3 (q, J = 271 Hz), 120.1 (q, J = 272 Hz), 18.55. IR (neat): ν = 2985, 2930, 1735, 1320, 1240, 1135, 1047, 708, 612 cm^{-1}. HRMS (ESI) m/z: calcd. for $C_{24}H_{17}F_6NO_4S_2Na$ [M + Na]$^+$: 584.0395, found: 584.0396.

■ ASSOCIATED CONTENT

Data Availability Statement

All the data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.4c00662.

Spectroscopic data of starting materials; copies of ^{1}H, ^{13}C{^{1}H}, ^{19}F NMR, ^{77}Se NMR spectra, and crystal data of the products (PDF)

Accession Codes

CCDC 2209544 and 2314583 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The

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Notes

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